

INAUGURAL CLASS • SECOND SEMESTER 2026

Special Topics in Computational Physics

Computational fluid dynamics: theory, numerical methods and validation.

Yachay Tech University • School of Physical Sciences and Nanotechnology

Physics • ECFN-PHY-1112 • Professional unit • In person

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The course at a glance

Structure of the course

144

total hours

48

contact hours

6

units

24

topics

Course description

Computational fluid dynamics as a current research topic, from the governing equations to the design, implementation and validation of solvers.

Prerequisites

No formal requirements; assumed background

Assumed background

- Vector calculus
- PDEs: hyperbolic, parabolic, elliptic
- Conservation of mass, momentum and energy
- Programming in Python, C++ or Julia
- Experience diagnosing failed runs



No prerequisites

None in the syllabus



No corequisites

None required



Recommended

Computational Physics I, or equivalent

Why computational fluid dynamics?

Governing equations and their solutions

I

Compact equations

The Navier–Stokes equations describe blood flow, aerofoils, ocean currents and the interstellar medium.

II

Few exact solutions

Analytic solutions are restricted to simple geometries. Turbulence has no general analytic description.

III

Numerical solution

A solver is a numerical experiment. Accuracy, stability and validation determine whether its output is meaningful.

Learning outcomes

What the course is designed to develop

1

Formulation of the physics

Conservation laws in integral and differential form.

2

Selection of numerical methods

Finite volumes for compressible flow; projection methods for incompressible flow.

3

Design and validation of solvers

Assessment of accuracy, stability and convergence against benchmarks.

4

Analysis of simulation data

Turbulence, boundary layers, instabilities, drag and lift.

Course structure

Six curricular units

1

Fluids and a first solver

Statics, control volumes, advection

2

Transport and stability

Burgers, CFL, finite volumes

3

Viscous flow

Navier–Stokes, vorticity, pressure

4

Incompressible solvers

Cavity, cylinder, validation

5

Boundary layers

Separation, drag, turbulence

6

Compressible flow

Shocks, instabilities, parallelism

1

UNIT 1

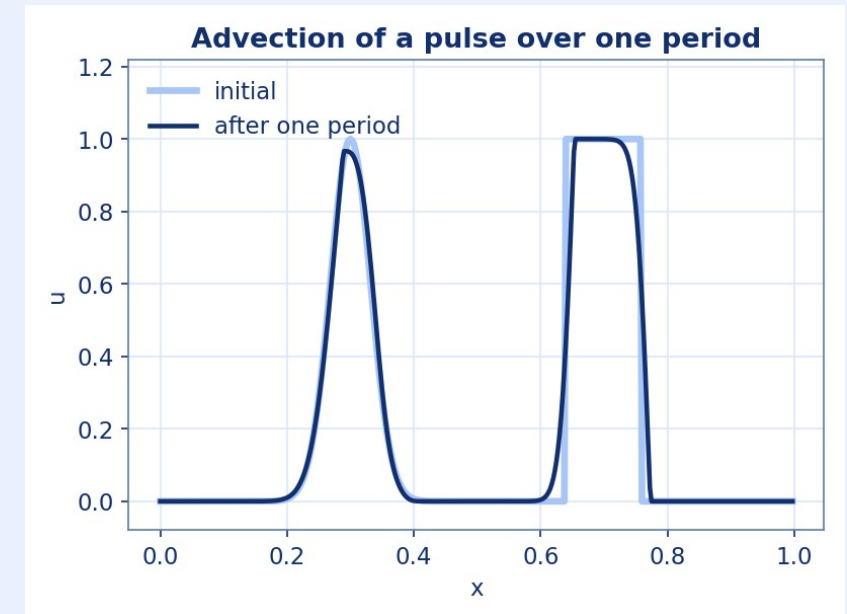
Fluids, conservation laws and a first solver

1 Fluid properties and the continuum hypothesis

2 Statics, buoyancy, hydrostatic forces

3 Control volumes and Bernoulli's equation

4 Grids, discretisation, advection



A pulse advected once around a periodic domain, showing numerical diffusion.

Own figure

This unit concludes with a working one-dimensional solver.

2

UNIT 2

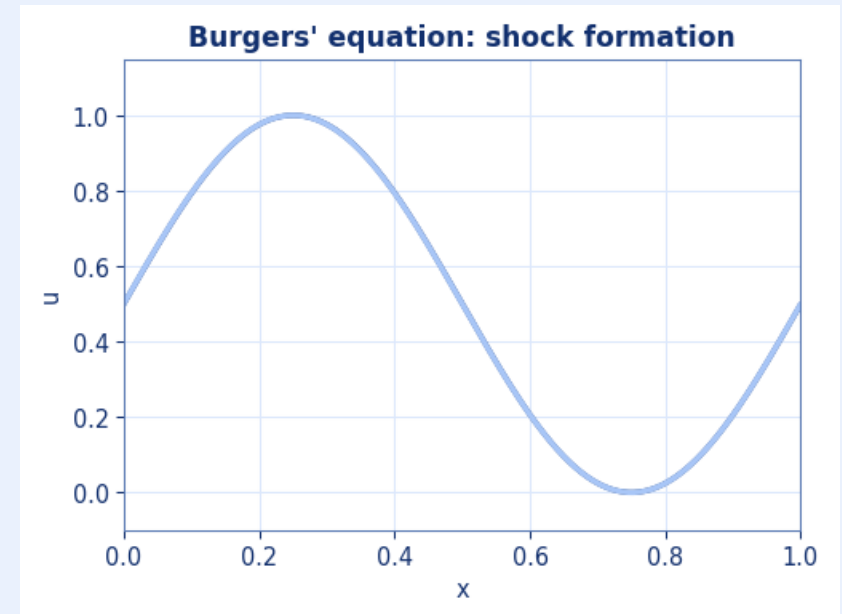
Kinematics, transport and numerical stability

1 Lagrangian and Eulerian descriptions

2 Burgers' equation: steepening and shocks

3 Diffusion, the CFL condition, stability

4 Finite volumes and conservative fluxes



Wave steepening and shock formation in Burgers' equation.

Own figure

Stability constraints and the origin of numerical instabilities.

3

UNIT 3

Differential analysis and viscous flow

1 Continuity and the Navier–Stokes equations

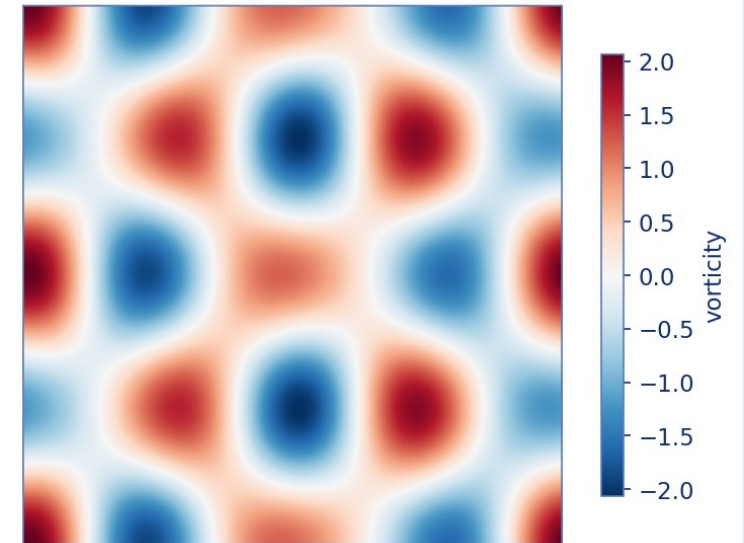
2 Couette, Poiseuille and potential flow

3 Vorticity–streamfunction formulation

4 Pressure–Poisson equation and projection

Pressure acts as the constraint that enforces incompressibility.

Vorticity of a decaying vortex array



Vorticity field of a decaying two-dimensional vortex array.

Own figure

4

UNIT 4

Incompressible solvers, similitude and validation

1

Lid-driven cavity and channel flow

2

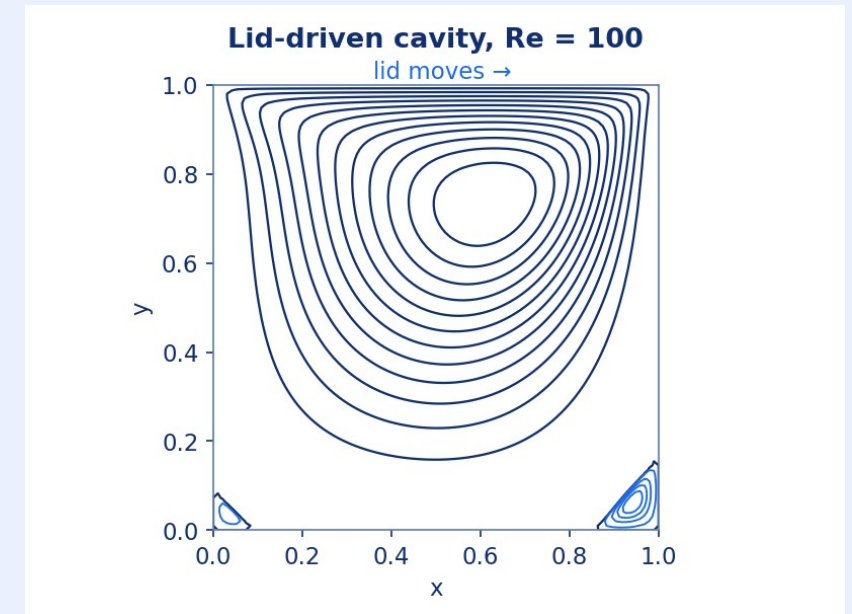
Cylinder wake and vortex shedding

3

Buckingham Pi theorem and similitude

4

Grid convergence and benchmark data



Streamlines in the lid-driven cavity benchmark at $Re = 100$.

Own figure

Verification assesses the numerics; validation assesses the model.

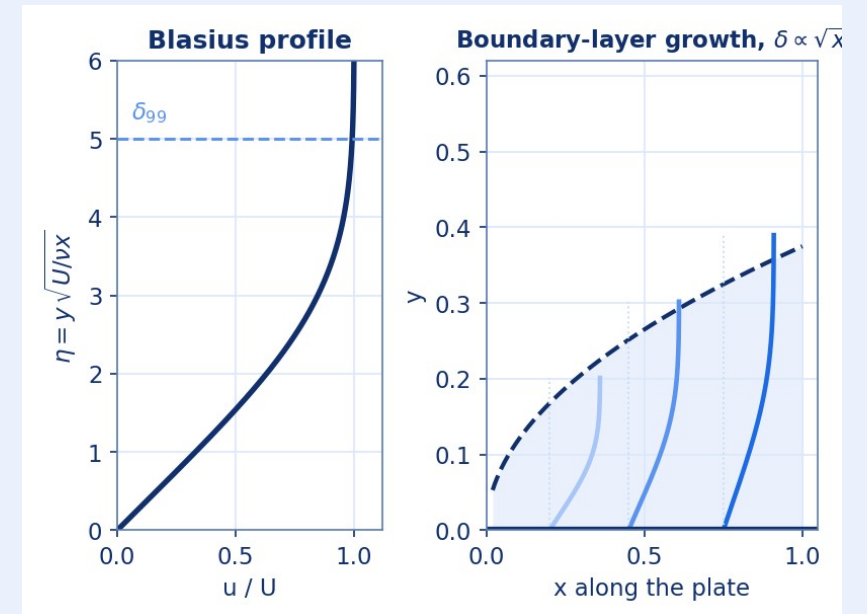
5

UNIT 5

Boundary layers and turbulence

- 1 Internal flow and the Moody chart
- 2 Boundary layers and flow separation
- 3 Drag, lift and aerofoil polars
- 4 The energy cascade and turbulence modelling

Resolution requirements and the limits of turbulence modelling.



The Blasius similarity profile and boundary-layer growth along a flat plate.

Own figure

6

UNIT 6

Compressible flow and instabilities

1

Mach number and nozzle flow

2

Riemann problems and Godunov schemes

3

Kelvin-Helmholtz and Rayleigh-Taylor

4

Parallel and mesh-free methods

Kelvin-Helmholtz roll-up of a shear layer



Passive tracer in a Kelvin-Helmholtz roll-up.

Own figure

Topics that connect directly to current research.

Teaching methodology

Course delivery and working tools

Course delivery

- Each unit pairs theory with solver implementation
- Solvers are tested, diagnosed and benchmarked
- Defences are written and closed-book
- Assessment covers the analysis, not only the output
- Bring a laptop if you can



Python

Main language



C++ and Julia

Where execution speed matters



Visualisation

Analysis of simulation data

Assessment

Two terms of the period, 50% each



■ Homework defences ■ Quizzes
■ Midterm exam ■ Final exam

25% **Homework defences**
Report and written defence

25% **Quizzes**
Physics, mathematics and programming

25% **Midterm exam**
First term of the period

25% **Final exam**
Second term of the period

Composition of each component

All defences are written and closed-book

25p Quizzes

- Closed-book, written
- Concepts, physics, mathematics, programming
- At least 4 during the semester

25p Homework defences

- 5p — complete, submitted report
- 20p — closed-book, written defence
- On your report and code • at least 2

25p Midterm exam

- 10p — quiz-like: closed-book, written
- 5p — complete, submitted notebook
- 10p — closed-book, written defence

25p Final exam

- 10p — quiz-like: closed-book, written
- 5p — complete, submitted notebook
- 10p — closed-book, written defence

Calendar

Second semester 2026

Classes

Every week

Tue 17:00–19:00 • Wed 17:00–19:00 • Thu 13:00–15:00

Tutoring

Every week

Tue 16:00–17:00 • Wed 16:00–17:00 • 2nd floor, Senescyt Building, or Zoom

30 Sep – 8 Oct

Midterm exam

First term of the period

7 – 10 Dec

Final exam

Second term of the period

Announced in class

Quizzes and defences

Once fixed, they are hard deadlines

Regulations

UITEY Academic Regulations

70

Attendance

70% minimum to pass. Absences justified within 5 days. (Art. 82–83)

6

Make-up exam

A final average of 4.8–5.9 qualifies. Passing raises the grade to 6.0. (Art. 61)

!

Academic integrity

No copying from other students, solution manuals, previous years' solutions, or AI chatbots.

=

Deadlines

Agreed in class; once fixed, they are hard deadlines.

Attendance and late submissions

Course policy

M

Attendance is recorded on Moodle

I record it myself after every class. Please check it regularly and let me know if there is an issue.

70

70% attendance is required to pass

As established by the Vicerrectorado of Yachay Tech and the University regulations.

5

Absences must be justified within 5 days

Official documentation only: the medical centre or Bienestar Universitario, as the regulations establish.

%

Late submissions

With appropriate justification, no penalty. Without it, a penalty of –1% per late hour applies.

Bibliography

Main and complementary sources

Main

Fluid Mechanics: Fundamentals and Applications

2014

Çengel & Cimbala • 3rd ed. • McGraw Hill

Fluid Mechanics

2011

Frank M. White • 7th ed. • McGraw Hill

Complementary

Fundamentals of Fluid Mechanics

2013

Munson, Okiishi, Huebsch & Rothmayer • 7th ed. • Wiley

Riemann Solvers and Numerical Methods for Fluid Dynamics

2009

Eleuterio F. Toro • 3rd ed. • Springer

Recommendations

How to approach the course

1

Submit your OWN work

Write code every week.

Steady weekly work is more effective than concentrated effort before a deadline.

2

Verify your results

Check conservation, refine the grid, and compare against exact solutions and benchmark data.

3

Build incrementally

Verify the one-dimensional case before extending to two dimensions, and keep each stage working.

Course materials

Scan the code or type the address



Course repository

github.com/wbandabarragan/computational-fluid-dynamics

Let's get started.



Bring a laptop if you can.

First session: fluid properties and the continuum hypothesis.

Lecturer

Wladimir E. Banda-Barragán

Tutoring

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or Zoom

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